MILITARY SPECIFICATION

SMOKE POT, FLOATING, SGF2, AN-M7A1; COMPONENTS FOR

This specification is approved for use within the Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers eight components for the AN-M7A1 smoke pot. The components are identified in table I.

20 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1168 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using: the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 1365

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DRAWINGS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

36-2-195 - Outer Cover Assy
36-2.196 - Cover, Lug
36-2-198 - Partition, Air
36-2-199 - Body
36-2-206 - Venturi-Fuel Chamber Assembly
36-2-211 - Ring, Clamp
36-2-217 - Container, Fuel Block
36-2-218 - Spacer
36-2-219 - Fuel Block Assembly

(Copies of standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3* REQUIREMENTS

3.1 Materials. All materials cited on the component drawings in table I and any subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

TABLE I. Component drawings

Component	Drawing
Outer cover assembly	36-2-195
Cover, lug	36-2-196
Partition, air	36-2-198
Body	36-2-199
Ring, clamp	36-2-211
Container, fuel block	36-2-217
Spacer	36-2-218
Fuel block assembly	36-2-219

3.2 Components. All components shall conform to the specifications and drawings listed on the drawings in table I and any subsidiary drawings.

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3.3 Manufacture and assembly. The components shall be manufactured and assembled in accordance with their respective drawings listed in table I.

3.4. Body leakage. The body shall not leak when tested as specified in 4.4.4.1.

3.5 Ring, clamp weld. The ring, clamp weld shall withstand a torque of no less than 100 inch-pounds and shall show *no* evidence of defects or damage when tested as specified in 4.4.4.2.

3.6 Fuel block assembly functioning. The fuel block assembly shall burn for no less than 8 minutes and no more than 13 minutes when tested as specified in 4.4.4.3.

3.7 Workmanship. All components shall be free of burrs, chips, sharp edges, cracks, surface defects, nonumiformities, dirt, grease, rust, corrosion products, and other foreign matter except as allowed in the applicable drawing. The cleaning method shall not be injurious to any of the parts.

3.8 First article inspection. Prior to the start of regular production, first article samples of components shall be produced in accordance with this specification for examination and test. Requirements for the submission of first article samples shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. *The* inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Samples. Five first article samples of each component shall be produced using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure. As determined by the Government, the first article samples may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection criteria. If any first article component sample fails to comply with any of the applicable requirements, the first article component sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any component sample to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the components of one interfix number, produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions. Each lot of components shall be identified and controlled in accordance with MIL-STD-1168. In addition, no more than one lot of each chemical ingredient shall be represented in each lot of fuel block assemblies.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with inspection level S-3 of MIL-STD-105. One hundred percent inspection shall be conducted for body leakage.

4.4.2.2 For destructive test. Sampling of fuel block assemblies for destructive testing shall be conducted in accordance with inspection level S-4 of MIL-STD-105.

4.4.3 Inspection procedure.

4,4.3.1 For examination and nondestructive tests. Sample components shall be examined in accordance with the classification of defects in 4.4.3.3 and MIL-STD-105. Any body that fails the body leakage test shall not be used in the manufacture of smoke pots.

4.4.3.2 For destructive tests. Fuel block assemblies shall be tested for functioning as specified in 4.4 .4.3 using an AQL of 1.0 percent defective.

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4.4.3.3 Classification of defects.

(a) Outer cover assembly (Drawing 36-2-195).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Diameter (11.120) incorrect
102	Dimension (.437) incorrect
103	Dimension (.22R) incorrect
104	Protective finish missing or incorrect
105	Workmanship not as specified (see 3.7)

(b) Cover, lug (Drawing 36-2-196).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Dimension (1.562) incorrect
102	Dimension (11.200) incorrect
103	Diameter (6.00) incorrect
104	Holes (.661) diameter incorrect
105	Holes (.719) diameter incorrect
106	Location (.719) diameter incorrect
107	Protective finish missing or incorrect
108	Workmanship not as specified (see 3.7)

(c) Partition, air (Drawing 36-2-198).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Diameter (11.250) incorrect
102	Diameter (10.81) incorrect
103	Diameter (4.00) incorrect
104	Hole (1.00) diameter incorrect
105	Location (1.00) diameter incorrect
106	Hole (.187) diameter incorrect
107	Location (.187) diameter incorrect
108	Hole (.125) diameter incorrect

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109	Location (.125) diameter incorrect
110	Dimension (.44) incorrect
111	Protective finish missing or incorrect
112	Workmanship not as specified (see 3.7)

(d) Body (Drawing 36-2-199).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	100 percent inspection	
101	Leakage	4.4.4.1
Major:	AQL 2.5 percent defective	
102	Inside diameter (11.250) incorrect	
103	Diameter (10.80) incorrect	
104	Dimension (13.62) incorrect	
105	Welding incorrect	
106	Protective finish missing or incorrect	
107	Workmanship not as specified (see 3.7)	

(e) Ring, clamp (Drawing 36-2-211).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 2.5 percent defective	
101	Weld incorrect	4.4.4.2
102	Components missing	
103	Stock material size incorrect	
104	Angle (60°) incorrect	
105	Outside diameter incorrect	
106	Break of sharp edges exceed (.010) max	
107	Gap (.375) incorrect	
108	Workmanship not as specified (see 3.7)	

(f) Container, fuel block (Drawing 36-2-217).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Outside diameter (8.323) incorrect
102	Inside diameter (8.125) incorrect

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103	Dimension (5.50) incorrect
104	Dimension (.12) incorrect
105	Hole diameter (.75) incorrect
106	Sealing material missing at seam joints
107	Workmanship not as specified (see 3.7)

(g) Spacer (Drawing 36-2-218).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Inside diameter incorrect
102	Wall thickness incorrect
103	Overlap dimension incorrect
104	Overlap not secured
105	Workmanship not as specified (see 3.7)

(h) Fuel block assembly (Drawing 36-2-219).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Diameter of fuel mixture incorrect
102	Height of fuel mixture incorrect
103	Diameter of ignition mixture incorrect
104	Height of ignition mixture incorrect
105	Diameter of hole incorrect
106	Depth of hole incorrect
107	Coating on top surface missing
108	Weight of fuel block (uninhibited) incorrect
109	Workmanship not as specified (see 3.7)

4.4.3.4 Inspection equipment. When special inspection equipment is required to perform examinations and tests, the contractor shall submit all inspection equipment designs for Government approval in accordance with the terms of the contract (see 6.2).

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Body leakage. Special inspection equipment is required for this test. Place the body in a suitable fixture that will seal all openings. Subject the body to an internal. air pressure of 5.0 ± 0.5 pounds per square inch gage through a suitable filling in the fixture. Thoroughly wipe the welded seam of the body with a leak detecting solution and observe for bubbles. The presence of bubbles shall be considered evidence of leakage.

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4.4.4.2 Ring, clamp weld. Torque the ring, clamp when the ring is assembled over a solid steel form with the bolt-nut torque at a minimum of 100 inch-pounds and visually inspect the weld for defects or damage.

4.4.4.3 Fuel block assembly functioning. Special inspection equipment is required for this test. Place the fuel block assembly in a fixture which simulates the venturi-fuel block chamber configuration (Drawing 36-2-206). Immerse the fixture to a depth of 4.5 ± 0.5 inches in water which is within 5°F of ambient temperature. Ignite the fuel block through the venturi tube of the fixture. Measure the time of burning.

5. PACKAGING

5.1 Packing and marking. The smoke pot components shall be packaged level B in accordance with MIL-STD-1169 unless otherwise specified by the contracting officer.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) First article requirements:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract
(2) Name and address of the test facility and shipping instructions when testing is performed by the Government
(3) Time required for the Government to notify the contractor whether to proceed with production

6.2 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to Commander, AMSMC-QAC-M (A), Aberdeen Proving Ground, MD 21010-5423 or as directed by the contract.

6.3 Subject term (key word) listing.

Fuel block
Smoke
Smoke pot

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1365-A088

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-51537(EA)	SMOKE POT, FLOATING, SGF2, AN-M7A1; COMPONENTS FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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INCH-POUND

MIL-S-51537(EA)
AMENDMENT 1
8 June 1989

MILITARY SPECIFICATION

SMOKE POT, FLOATING, SGF2, AN-M7A1; COMPONENTS FOR

This amendment forms a part of MIL-S-51537(EA). dated 16 October 1987, and is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.6, line 2: Delete “8 minutes" and substitute “9 minutes”.

PAGE 7

4.4.3.3(h): Delete defect 108 in its entirety

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1365-A097

AMSC N/A

FSC 1365

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NOTICE OF INACTIVATION FOR NEW DESIGN

INCH-POUND

MIL-S-51537(EA)
NOTICE 1
18 January 1996

MILITARY SPECIFICATION

SMOKE POT, FLOATING, SGF2, AN-M7A1; COMPONENTS FOR

This notice should be filed is front of MIL-S-51537(EA), dated 16 October 1987, with Amendment 1, dated 8 June 1989.

MIL-S-51537(EA) is inactive for new design and is no longer used except for replacement purposes.

Preparing activity;

Army - EA

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